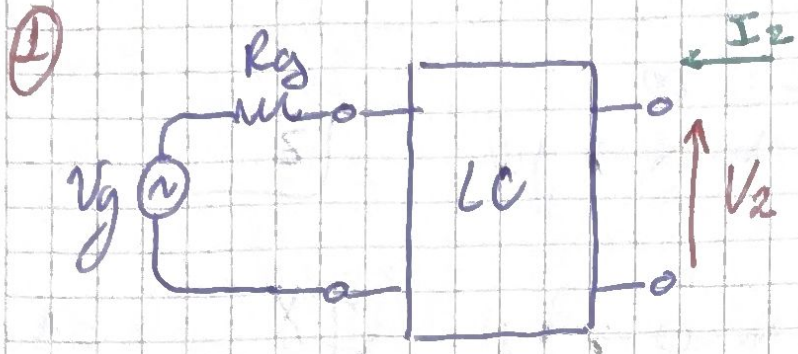
Lucas Gallero

HOJA Nº 1

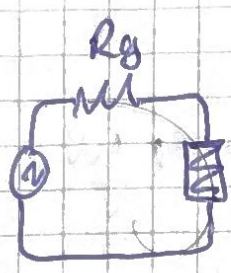
FECHA

TS9 - TC2

① 

$$T(s) = \frac{V_2}{V_g} = K\,\frac{s(s^2+1/16)}{s^3+2s^2+2s+1}$$

$$\left.\frac{V_2}{V_g}\right|_{I_2=0}$$



$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} \qquad \frac{V_1}{V_g} = \frac{Z_{11}}{Z_{11}+R_g}$$

$$\left.\frac{V_2}{V_g}\right|_{I_2=0} = \left.\frac{V_1}{V_g}\right|_{I_2=0} \cdot \left.\frac{V_2}{V_1}\right|_{I_2=0} = \frac{Z_{11}}{R_g+Z_{11}} \cdot \frac{Z_{21}}{Z_{11}}$$

$$\boxed{\left.\frac{V_2}{V_g}\right|_{I_2=0} = \frac{Z_{21}}{R_g+Z_{11}}}$$

Normalizo → $R_g = 1$.

$$\boxed{\left.\frac{V_2}{V_g}\right|_{I_2=0} = \frac{Z_{21}}{1+Z_{11}}} \Rightarrow K\,\frac{s(s^2+1/16)}{s^3+2s^2+2s+1}.$$

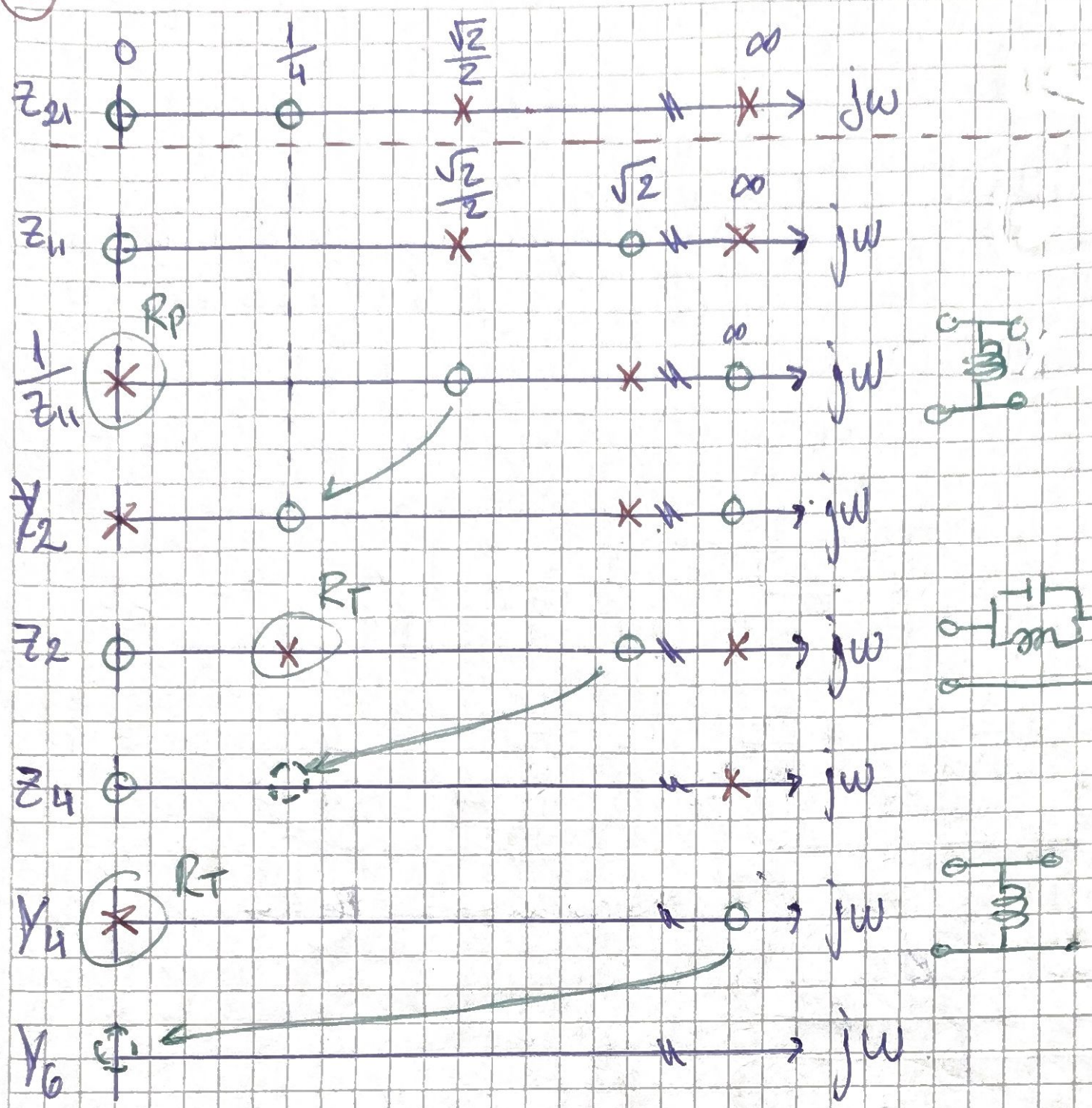
Z_{21} es impar → Divido numerador y denominador por la parte par de $Q(s)$ → $2s^2+1$

$$T(s) = \frac{\dfrac{K(s^2+1/16)s}{2s^2+1}}{1+\dfrac{s^3+2s}{2s^2+1}} \rightarrow \boxed{Z_{21} = K\cdot\frac{s(s^2+1/16)}{2s^2+1}}$$

$$T(s) = \frac{Z_{21}}{1+Z_{11}} \qquad \boxed{Z_{11} = \frac{s(s^2+2)}{2s^2+1}}$$

NOTA

a)

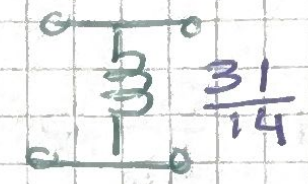


	0	∞
Z	$\frac{1}{sC}$	sL
Y	$\frac{1}{sL}$	sC

NOTA

$$Z_{11} = \frac{S(S^2+2)}{2S^2+1} \rightarrow \frac{1}{Z_{11}} = \frac{2S^2+1}{S(S^2+2)}$$

$$K_0 = \lim_{S^2 \to -\frac{1}{1}} \cdot S \cdot \frac{2S^2+1}{S(S^2+2)} = \frac{14}{31} \rightarrow$$



$$Y_2 = \frac{2S^2+1}{S(S^2+2)} - \frac{14/31}{S} = \frac{2S^2+1 - 14/31(S^2+2)}{S(S^2+2)}$$

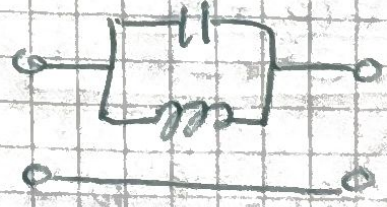
$$Y_2 = \frac{48/31\, S^2 + 3/31}{S(S^2+2)} = \frac{48/31\left(S^2 + \frac{1}{16}\right)}{S(S^2+2)}$$

$$Z_2 = 31/48 \; \frac{S(S^2+2)}{S^2 + 1/16}$$

$$2K_i = \lim_{S^2 \to -\frac{1}{16}} \cdot \frac{(S^2+1/16)}{S} \cdot \frac{31}{48} \cdot \frac{S(S^2+2)}{S^2+1/16} = \frac{961}{768}$$

$$\frac{2K_i S}{S^2+w^2} = \frac{1}{\frac{S^2}{2K_i S} + \frac{w^2}{S 2K_i}} = \frac{1}{\underbrace{\left(\frac{1}{2K_i}\right)}_{C} S + \frac{1}{S\underbrace{\left(\frac{2K_i}{w^2}\right)}_{L}}}$$

$$\boxed{C = \frac{768}{961}} \quad \text{y} \quad \boxed{L = \frac{961}{48}}$$



NOTA

$$Z_4 = \frac{31}{48}\,\frac{S(S^2+2)}{S^2+1/16} - \frac{961}{768}\cdot\frac{S}{S^2+1/16} = \frac{S^3+2S-\frac{961}{768}\cdot\frac{48}{31}S}{\frac{48}{31}\,S^2+1/16}$$

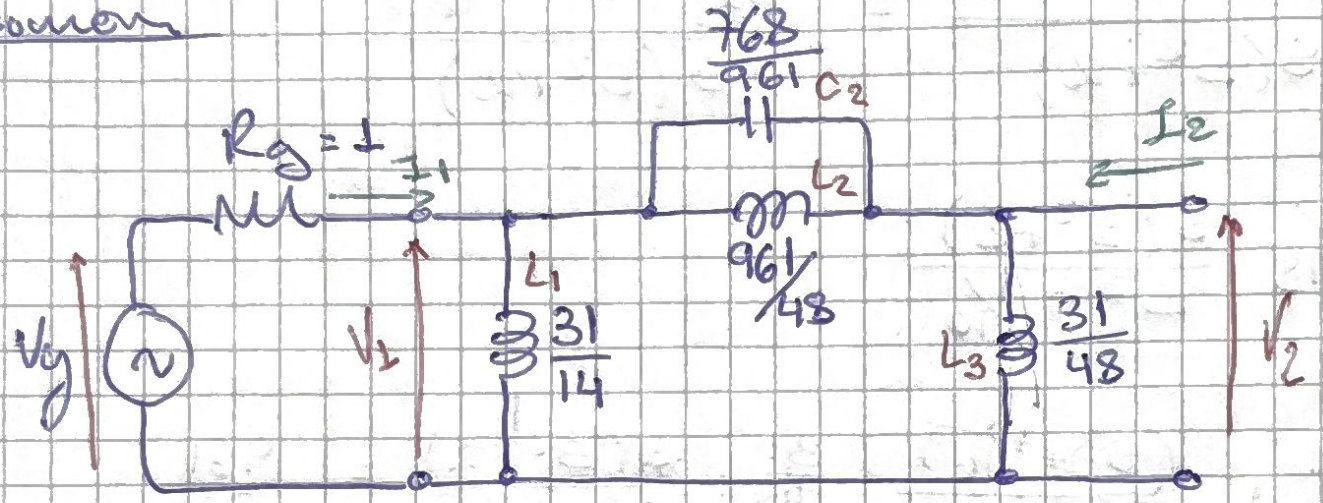
$$Z_4 = \frac{31}{48}\,\frac{S^3+\frac{1}{16}S}{S^2+1/16} = \frac{31}{48}\,\frac{S(S^2+1/16)}{(S^2+1/16)}$$

$$Z_4 = \frac{31}{48}S$$

$$Y_4 = \frac{48}{31}\cdot\frac{1}{S} \rightsquigarrow \boxed{L = \frac{31}{48}H}$$



Resonem



(b)

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$$

$$\boxed{Z_{21} = \frac{k\,S(S^2+1/16)}{2S^2+1}}$$

$$V_2 = I\cdot Z_{L_3}$$

$$I = I_1\cdot\frac{Z_{L_1}}{Z_4+Z_p+Z_{L_3}}$$

$$\begin{cases} Z_{L_3} = SL_3 = 31/48\,S \\ Z_{L_1} = SL_1 = 31/14\,S \\ Z_p = \dfrac{961}{768}\,\dfrac{S}{S^2+1/16} \end{cases}$$

$$V_2 = I_1\,\frac{Z_{L_3}Z_{L_1}}{Z_4+Z_p+Z_{L_3}}$$

$$\boxed{\frac{V_2}{I_1} = \frac{Z_{L_3}Z_{L_1}}{Z_4+Z_p+Z_{L_3}}}$$

NOTA

$$Z_{21} = \frac{V_2}{I_1} = \frac{\frac{31}{14}S \cdot \frac{31}{48}S}{\frac{31}{14}S + \frac{31}{48}S + \frac{961}{768}\frac{S}{S^2+1/16}}$$

$$Z_{21} = \frac{\frac{961}{672}S}{\frac{961}{336}S + \frac{961}{768}\frac{S}{S^2+1/16}} = \frac{\frac{672/1}{672}S}{\frac{672\,1}{336} + \frac{672\,1}{768}\frac{1}{S^2+1/16}}$$

$$Z_{21} = \frac{S}{2 + \frac{7}{8}\cdot\frac{1}{S^2+1/16}} = \frac{S}{\frac{2(S^2+1/16)+7/8}{(S^2+1/16)}}$$

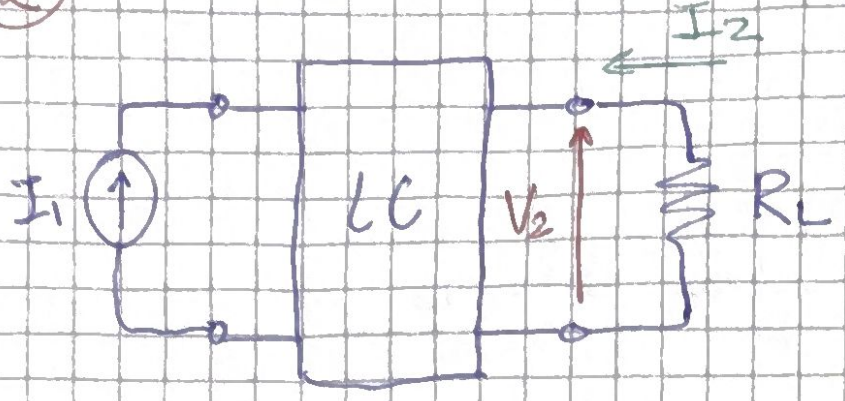
$$Z_{21} = \frac{S(S^2+1/16)}{2S^2+1}$$

$$Z_{21} = \frac{k\cdot S(S^2+1/16)}{2S^2+1}$$

$$k = 1$$

Papel de fibra de caña de azúcar.

(2)



$$T(s) = \frac{V_2}{I_1} = k \cdot \frac{(s^2+9)}{s^3+2s^2+2s+1}$$

$$V_2 = Z_{21} I_1 + Z_{22} \cdot I_2 \qquad \triangleright\ I_2 = -\frac{V_2}{R_L}$$

$$V_2 = Z_{21} I_1 + Z_{22}\left(-\frac{V_2}{R_L}\right)$$

$$-Z_{21} I_1 = -V_2 - Z_{22} \cdot \frac{V_2}{R_L}$$

$$Z_{21} I_1 = \left(1 + \frac{Z_{22}}{R_L}\right) V_2$$

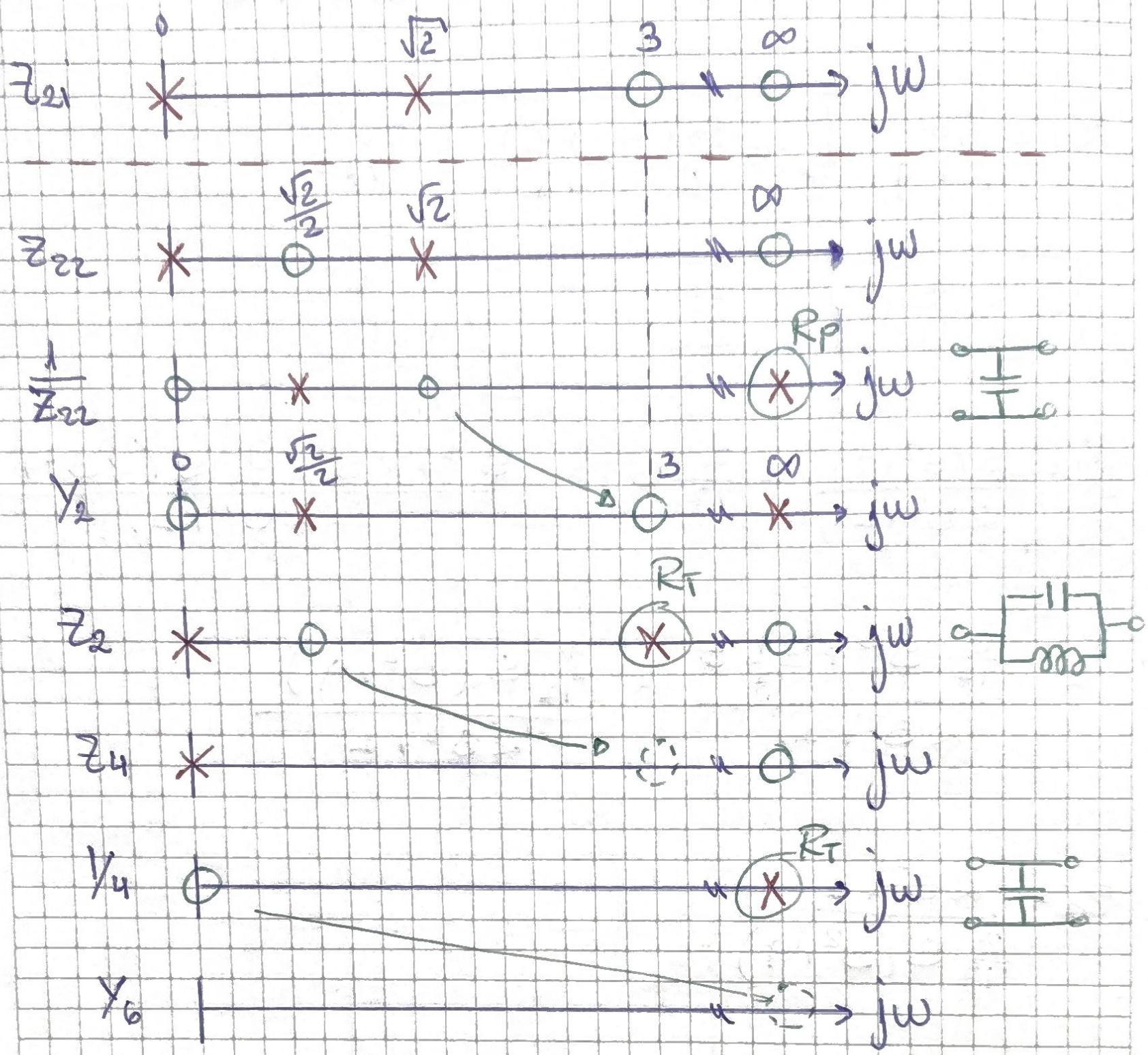
$$\boxed{\frac{V_2}{I_1} = \frac{Z_{21}}{1 + Z_{22}/R_L}}$$

$R_L = 1 \rightarrow$ NORMALIZADO

$$\boxed{\frac{V_2}{I_1} = \frac{Z_{21}}{1 + Z_{22}}} \rightarrow k \cdot \frac{(s^2+9)}{s^3+2s^2+2s+1}$$

- Como el numerador es PAR, multiplico y divido por la parte impar de $\Theta_+(s)$

$$\frac{V_2}{I_1} = \frac{\frac{k(s^2+9)}{s(s^2+2)}}{1 + \frac{2s^2+1}{s(s^2+2)}} \Rightarrow \begin{cases} Z_{21} = \dfrac{k(s^2+9)}{s(s^2+2)} \\ Z_{22} = \dfrac{2s^2+1}{s(s^2+2)} \end{cases}$$



	0	∞
Z	$\frac{1}{sC}$	sL
Y	$\frac{1}{sL}$	sC

Papel de fibra de caña de azúcar.

$$Z_{22} = \frac{2S^2+1}{S(S^2+2)} \longrightarrow \frac{1}{Z_{22}} = \frac{S(S^2+2)}{2S^2+1}$$

$$K_\infty = \lim_{S^2 \to -9} \frac{1}{S} \cdot \frac{S(S^2+2)}{2S^2+1} = \frac{7}{17}$$

$$\boxed{C_1 = \frac{7}{17} F}$$



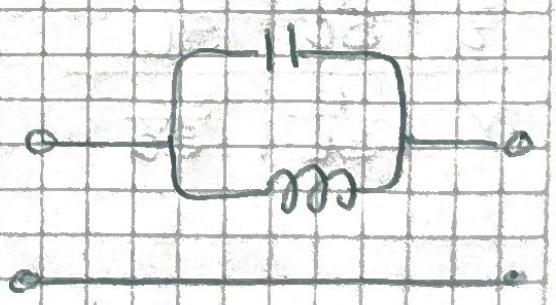
$$Y_2 = \frac{S(S^2+2)}{2S^2+1} - \frac{7}{17}S = \frac{S^3+2S-\frac{7}{17}S(2S^2+1)}{2S^2+1}$$

$$Y_2 = \frac{\frac{3}{17}S^3 + \frac{27}{17}S}{2S^2+1} = \frac{\frac{3}{17}S(S^2+9)}{2S^2+1}$$

$$Z_2 = \frac{2S^2+1}{\frac{3}{17}S(S^2+9)}$$

$$2k_1 = \lim_{S^2 \to -9} \frac{S^2+9}{S} \cdot \frac{17}{3} \cdot \frac{2S^2+1}{S(S^2+9)} = \frac{289}{27}$$

$$\frac{2k_1 S}{S^2+\omega^2} = \frac{1}{\frac{1}{2k}S + \frac{1}{S\left(\frac{2k_1}{\omega^2}\right)}}$$



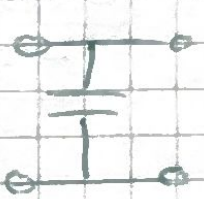
$$\boxed{C_2 = \frac{27}{289} F} \qquad \boxed{L_2 = \frac{289}{243}}$$

$$Z_4 = \frac{2S^2+1}{\frac{3}{17}S(S^2+9)} - \frac{\frac{289}{27}S}{S^2+9}$$

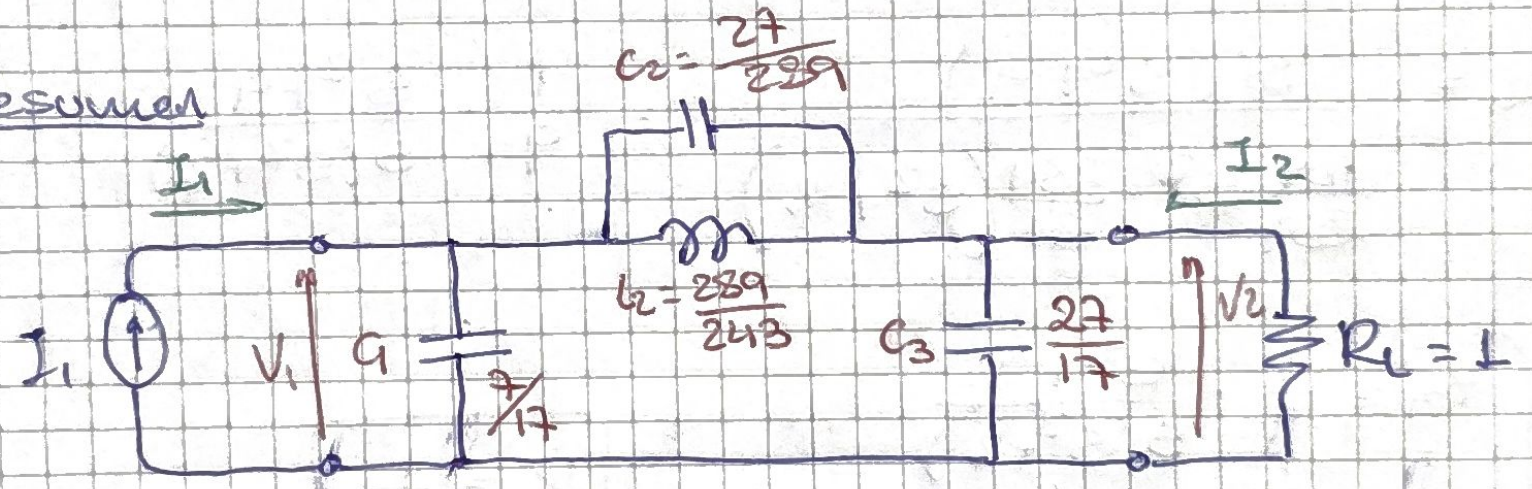
$$Z_4 = \frac{2S^2+1-\frac{289}{27}S\left(\frac{3}{17}S\right)}{\frac{3}{17}S(S^2+9)} = \frac{\frac{1}{9}S^2+1}{\frac{3}{17}S(S^2+9)}$$

$$Z_4 = \frac{\frac{1}{9}(\cancel{S^2+9})}{\frac{3}{17}S(\cancel{S^2+9})} = \frac{17}{27}\cdot\frac{1}{S}$$

$$Y_4 = \frac{27}{17}\cdot S \longrightarrow \boxed{C_3 = \frac{27}{17}\,F}$$



Resumen



(b)

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} = \frac{Z_{C_1}\cdot Z_{C_3}}{Z_{C_1}+Z_{C_3}+Z_p} = \frac{\frac{1}{SC_1}\cdot\frac{1}{SC_3}}{\frac{1}{SC_1}+\frac{1}{SC_3}+\frac{289}{27}\frac{S}{S^2+9}}$$

$$Z_{21} = \frac{\frac{1}{S\frac{7}{17}}\cdot\frac{1}{S\frac{27}{17}}}{\frac{1}{S\frac{7}{17}}+\frac{1}{S\frac{27}{17}}+\frac{289}{27}\frac{S}{S^2+9}}$$

Papel de fibra de caña de azúcar.

$$z_{21} = \frac{\frac{289}{189}\cdot\frac{1}{s^2}}{\frac{17/7}{s}+\frac{17/27}{s}+\frac{289}{27}\cdot\frac{s}{s^2+9}}$$

$$z_{21} = \frac{\frac{289}{189}\cdot\frac{1}{s^2}}{\frac{\frac{578}{189}}{s}+\frac{\frac{289}{27}s}{s^2+9}}$$

$$z_{21} = \frac{\frac{289}{189}\cdot\frac{1}{s^2}}{\frac{\frac{578}{189}(s^2+9)+\frac{289}{27}s^2}{s(s^2+9)}} = \frac{\frac{289}{189}\cdot\frac{1}{s^2}}{\frac{\frac{289}{21}s^2+\frac{578}{21}}{s(s^2+9)}}$$

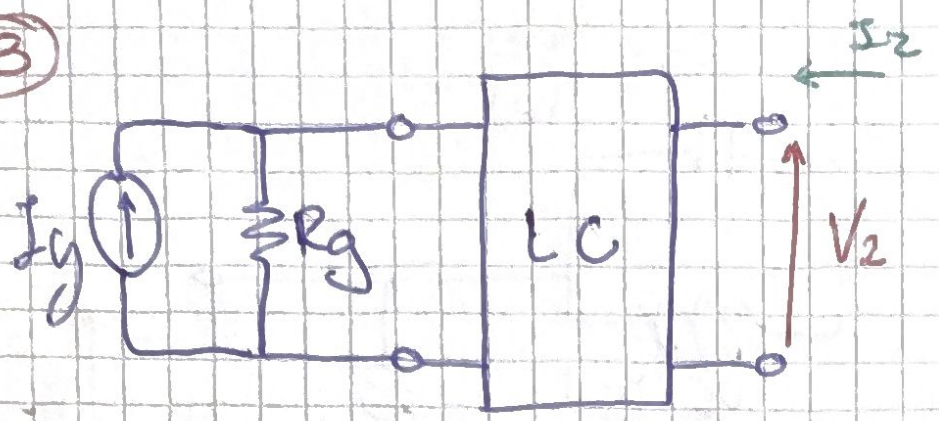
$$z_{21} = \frac{\cancel{8}(s^2+9)}{\frac{189}{289}\cancel{s^2}\cdot\frac{289}{21}(s^2+2)} = \frac{1}{9}\cdot\frac{s^2+9}{s(s^2+2)}$$

$$\begin{cases} z_{21} = \frac{1}{9}\cdot\frac{s^2+9}{s(s^2+2)} \\ z_{21} = \frac{k(s^2+9)}{s(s^2+2)} \end{cases} \Rightarrow \boxed{k=\frac{1}{9}}$$

HOJA Nº 6

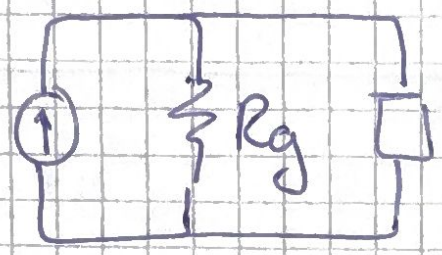
FECHA

(3)



$$T(S) = \frac{V_2}{I_g} = k\,\frac{S(S^2+1/16)}{S^3+2S^2+2S+1}$$

$$\left.\frac{V_2}{I_g}\right|_{I_2=0}$$



$$I_1 = I_g \cdot \frac{R_g}{R_g + Z_u}$$

$$\frac{V_2}{I_1} = Z_{21}$$

$$\frac{V_2}{I_g} = \frac{I_1}{I_g}\cdot\frac{V_2}{I_1} = \frac{R_g}{R_g+Z_u}\cdot Z_{21}$$

$$\boxed{\frac{V_2}{I_g} = \frac{Z_{21}}{1+Z_u/R_g}}$$

Normalizo $R_g = 1$.

$$\boxed{\frac{V_2}{I_g} = \frac{Z_{21}}{1+Z_u}} \Rightarrow \frac{k\,S(S^2+1/16)}{S^3+2S^2+2S+1}$$

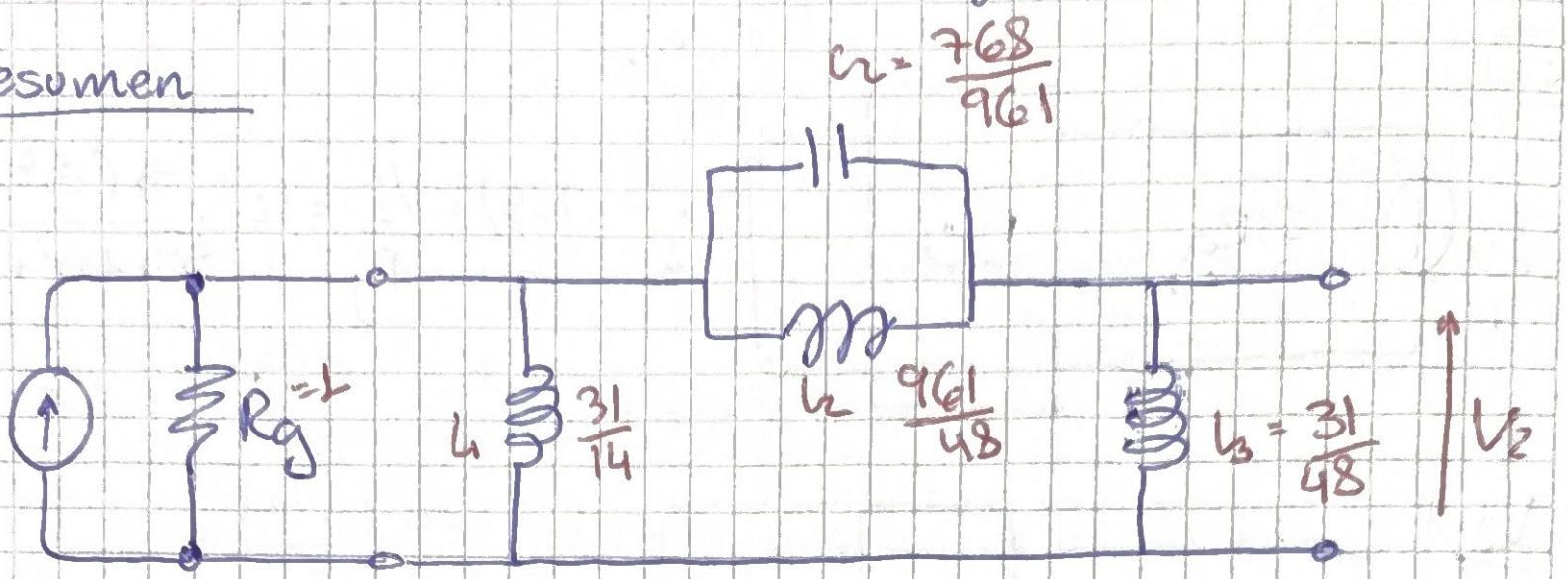
$$T(S) = \frac{\dfrac{k(S^2+1/16)\,S}{2S^2+1}}{1+\dfrac{S^3+2S}{2S^2+1}}$$

$$\boxed{Z_{21} = k\cdot\frac{S(S^2+1/16)}{2S^2+1}} \qquad \boxed{Z_{11} = \frac{S(S^2+2)}{2S^2+1}}$$

NOTA

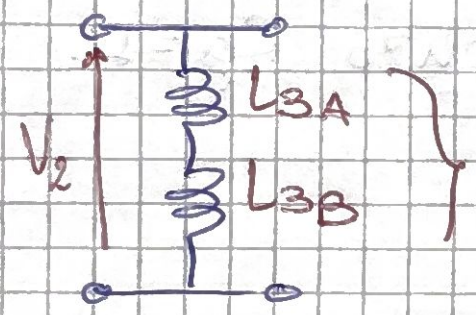
- las Remociones y la síntesis es igual al punto 1.

Resumen



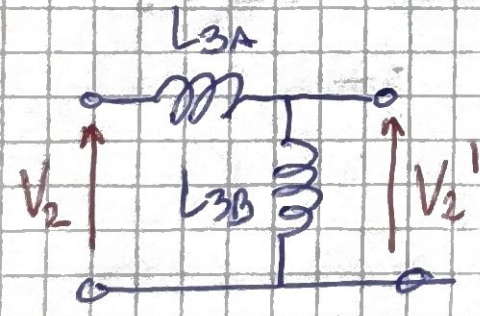
(b) Como necesito un k para tener -6dB en la Banda de paso.

Divido la inductancia L_3 en 2 inductores en serie



$$L_{3A} + L_{3B} = L_3 = \frac{31}{48}$$

- Tomo la salida desde L_{3B}



$$V_2' = V_2 \cdot \frac{SL_{3B}}{SL_{3B} + SL_{3A}}$$

$$\frac{V_2'}{V_2} = \frac{L_{3B}}{L_{3B} + L_{3A}} = \alpha$$

$$\boxed{V_2' = \alpha V_2}$$

NOTA

$$\frac{V_2}{Ig} = \frac{S(S^2+1/16)}{S^3+2S^2+2S+1} \qquad \frac{V_2'}{Ig} = \alpha \frac{V_2}{Ig}$$

$$\boxed{\frac{V_2'}{Ig} = \alpha \cdot \frac{S(S^2+1/16)}{S^3+2S^2+2S+1}}$$

$$\boxed{\alpha = k = \frac{L_{3B}}{L_{3A}+L_{3B}}}$$

$$L_{3A} + L_{3B} = \frac{31}{48} = L_3$$

$$k \overbrace{(L_{3A}+L_{3B})}^{L_3} = L_{3B}$$

$$\boxed{k \cdot L_3 = L_{3B}}$$

$$L_{3A} + L_{3B} = L_3$$

$$L_{3A} = L_3 - L_{3B} = L_3 - kL_3 = (1-k)L_3$$

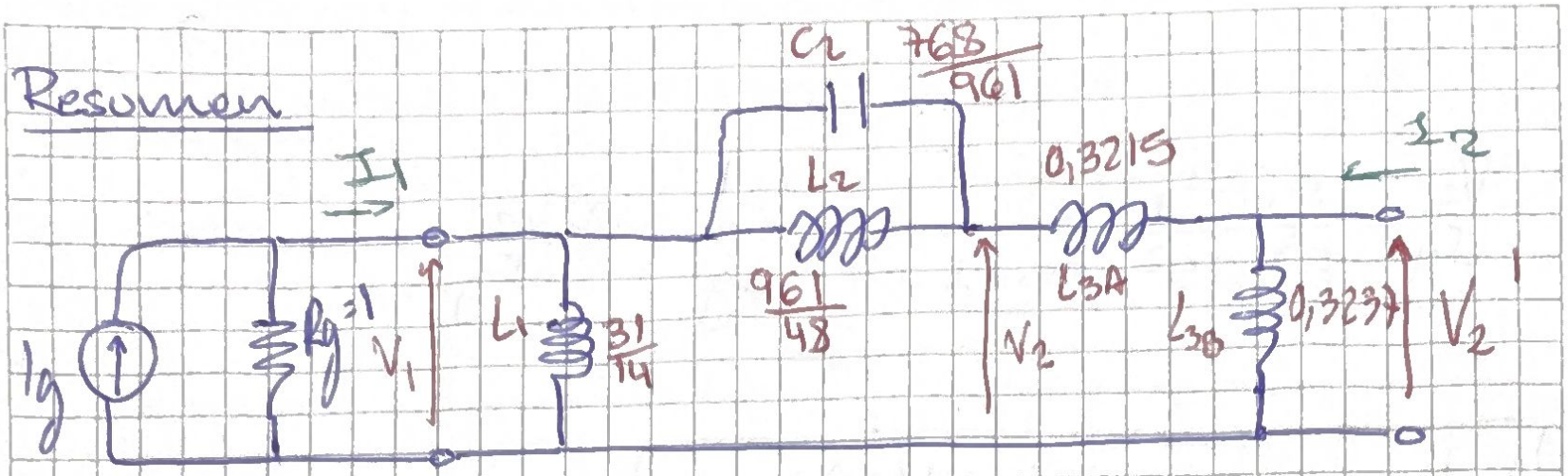
$$\boxed{L_{3A} = (1-k)L_3}$$

$$\boxed{k = 10^{-\frac{6}{20}} = 0,501187}$$

$$\boxed{L_{3A} = 0,3215} \qquad \boxed{L_{3B} = 0,3237}$$

NOTA

Resumen



NOTA